AP PRECALCULUS

Unit 1 • Lesson 1.4 practice • Answer key

Lesson 1.4 • Polynomial Functions and Rates of Change

5 multiple-choice questions and 1 free-response question • Answer key with explanations

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

What is the greatest possible number of local extrema of a polynomial of degree 5?

(A) 5
(B) 4
(C) 3
(D) 6

Solution

Answer: (B). A degree-n polynomial has at most $n-1$ local extrema.

Question 2

NO CALCULATOR

A polynomial p is increasing on $(-\infty, -2)$, decreasing on $(-2, 3)$ and increasing on $(3, \infty)$. Where does p have a local maximum?

(A) $x = 3$
(B) $x = 0$
(C) $x = -2$
(D) Nowhere

Solution

Answer: (C). p changes from increasing to decreasing at $x = -2$.

Question 3

NO CALCULATOR

What is the leading coefficient of $p(x) = 2x^3(x-1)^2 - x^5$?

(A) 2
(B) -1
(C) 1
(D) 3

Solution

Answer: (C). $2x^3(x^2 - 2x + 1) - x^5 = x^5 - 4x^4 + 2x^3$, so the leading coefficient is 1.

Question 4

NO CALCULATOR

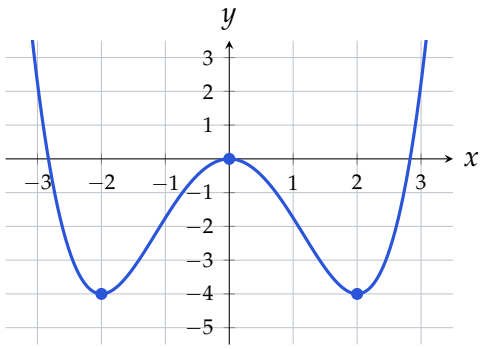
A point of inflection on the graph of a polynomial occurs where

(A) the rate of change changes from increasing to decreasing, or the reverse.
(B) the function changes from increasing to decreasing.
(C) the graph crosses the x-axis.
(D) the function has a local minimum.

Solution
Answer: (A). A point of inflection is a change in concavity, that is, in the behavior of the rate of change.

Question 5 (graph) NO CALCULATOR

The graph of the polynomial $p(x) = \frac{1}{4}x^4 - 2x^2$ is shown. What is the absolute minimum value of p?



(A) 0
(B) -2
(C) -5
(D) -4

Solution
Answer: (D). The lowest points of the graph are the two local minima $(-2, -4)$ and $(2, -4)$; the graph rises without bound at both ends, so the absolute minimum value is -4.

Question 6 (Free response) NO CALCULATOR

The polynomial $p(x) = x^3 - 6x^2 + 9x$ has a local maximum at $(1, 4)$, a local minimum at $(3, 0)$ and a point of inflection at $(2, 2)$.

(a) On which intervals is p increasing? decreasing?
(b) Find the absolute maximum and minimum values of p on $[0, 4]$.
(c) On which intervals is the graph of p concave up? concave down?

Solution
(a) Increasing on $(-\infty, 1)$ and $(3, \infty)$; decreasing on $(1, 3)$.
(b) Compare the local extrema with the endpoints: $p(0) = 0$, $p(1) = 4$, $p(3) = 0$, $p(4) = 64 - 96 + 36 = 4$. Absolute maximum 4 (at $x = 1$ and $x = 4$); absolute minimum 0 (at $x = 0$ and $x = 3$).
(c) The leading coefficient is positive, so the graph is concave down on $(-\infty, 2)$ and concave up on $(2, \infty)$.